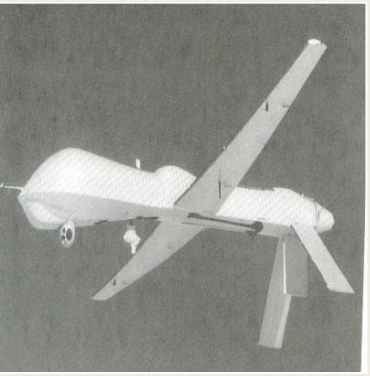

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

REMOTE SENSING PLATFORMS











ACQUISITION PLATFORMS

- Vehicle or carrier of remote sensing sensors
  - Airborne
  - Spaceborne
  - Ground –based- ladder trucks, Cherry Pickers

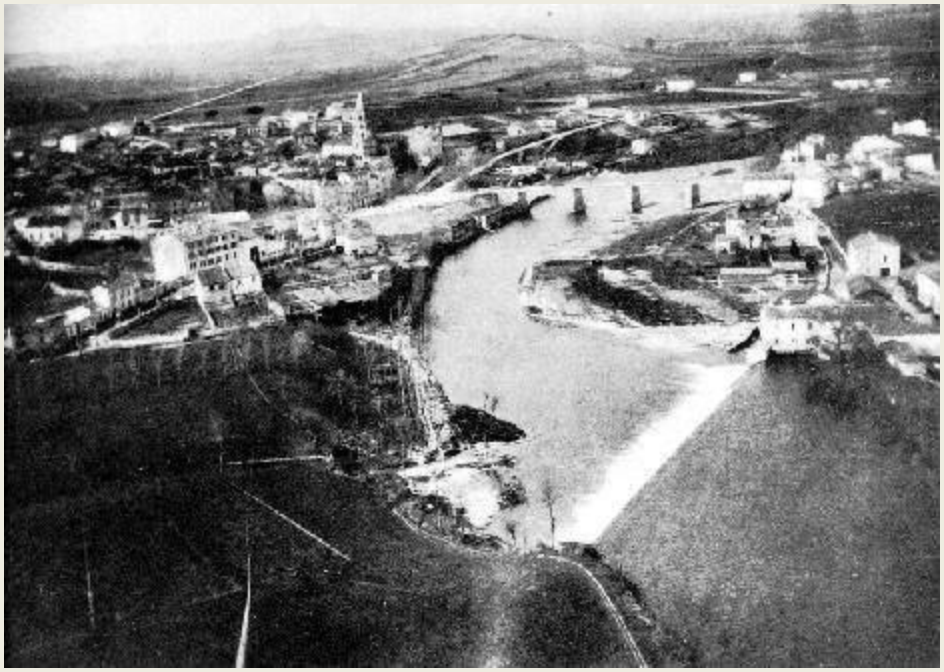
First Aerial Photograph- 1858



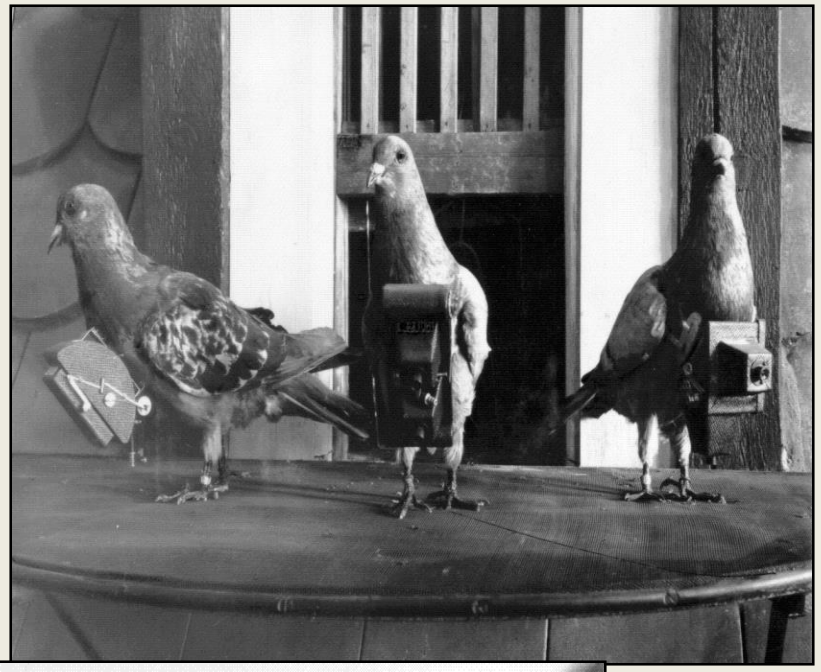


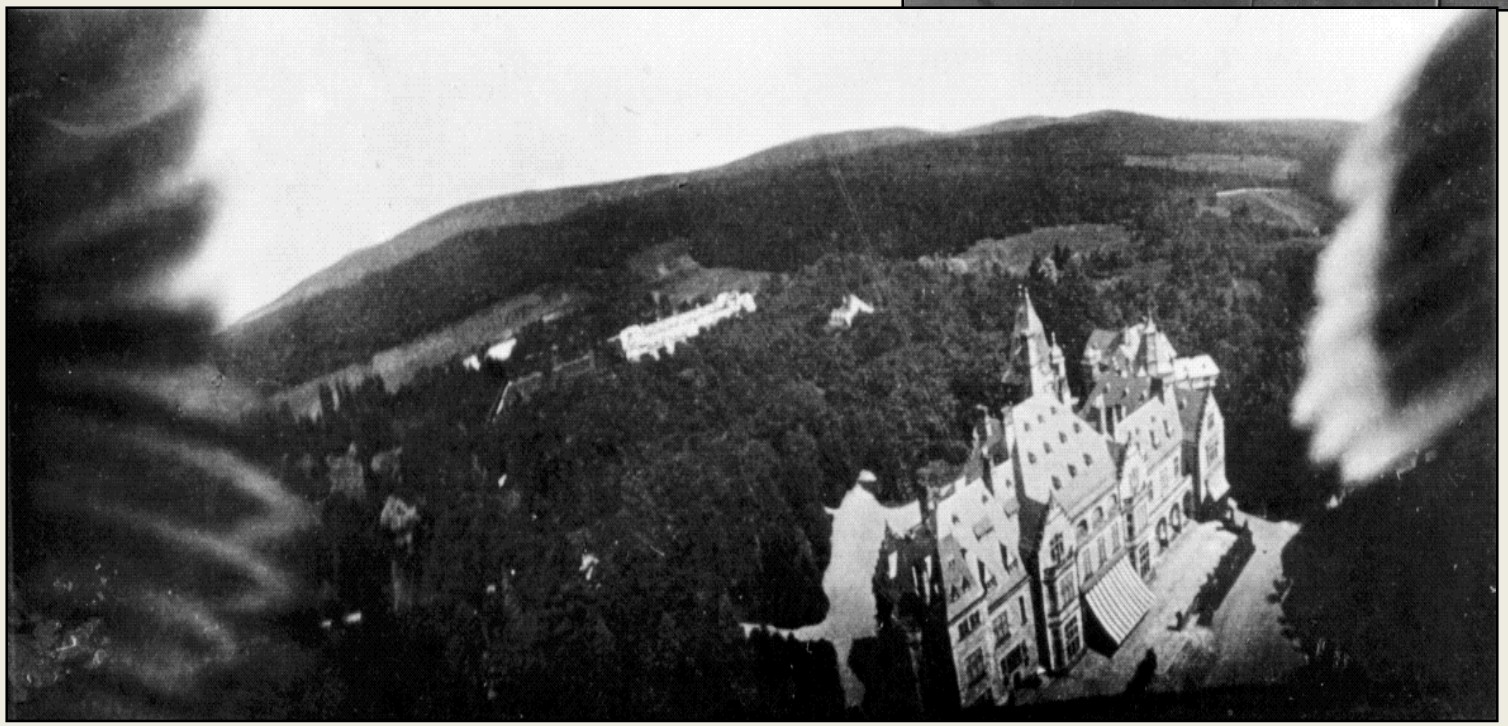
First aerial photographs using a kite in May 1888





Pigeons-1903





Aerial Photographs



SELECTION OF PLATFORMS

- Purpose
  - Coverage
    - Less or large area
    - Single or repeated
  - Orbits
  - Atmospheric conditions depending on the altitude

AIRCRAFTS

- First widespread use world war-1 for military intelligence
- A well developed and refined technique
- Currently air crafts flown with several type of sensor
- Flexibility of operation over any desired area at any time

AIRCRAFTS

- Most easily available RS data
- Data of higher resolution
- Instrument can be tuned and repaired
- Image can be taken with any solar illumination angle

AIRCRAFTS

- **Limitations –**
  - **Obtaining permission**
  - **Coordination of pilot and photographer**
  - **Smaller area coverage**
  - **Weather Dependence**
  - **Higher cost**
  - **Possibility of systematic distortions (roll, pitch, yaw)**


NASA
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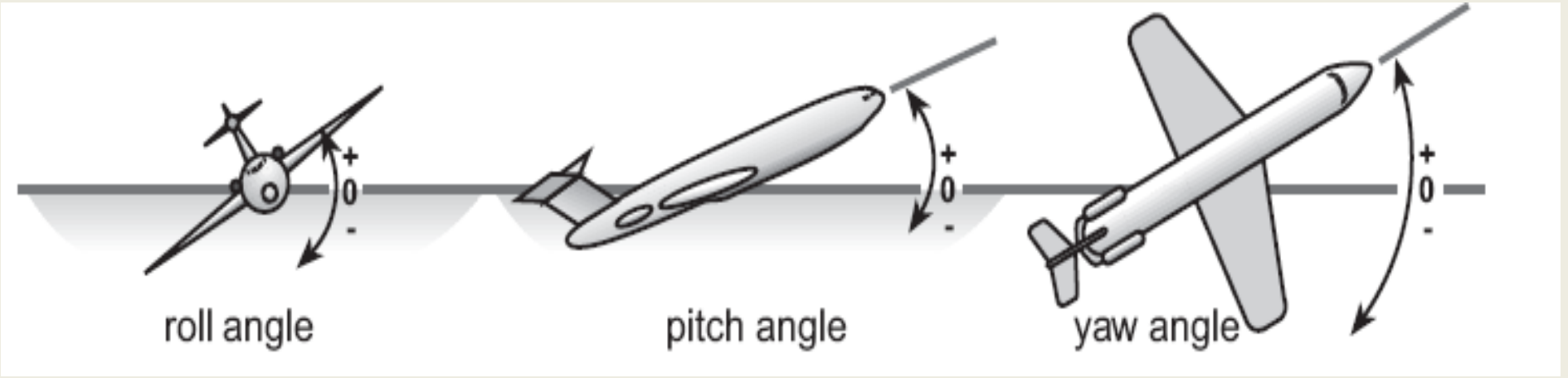


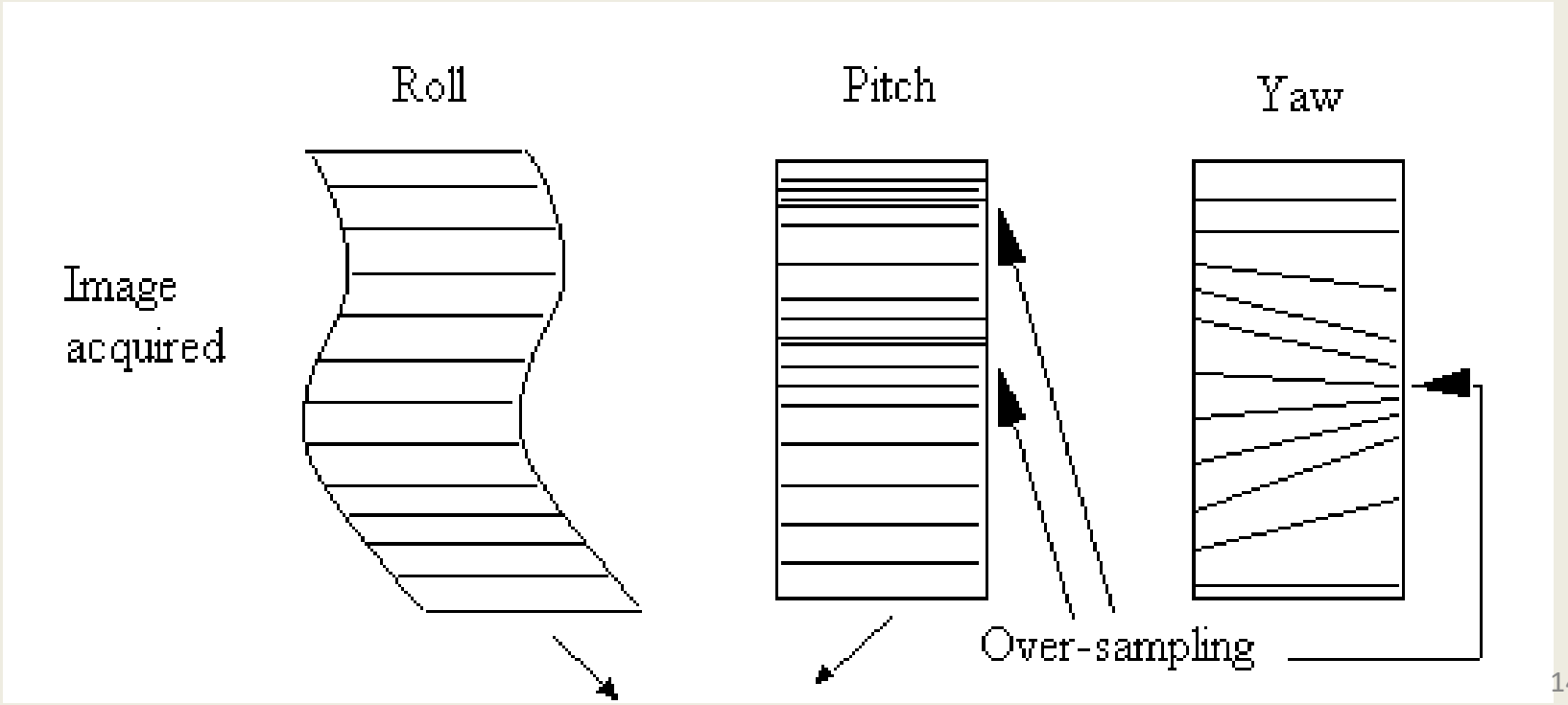

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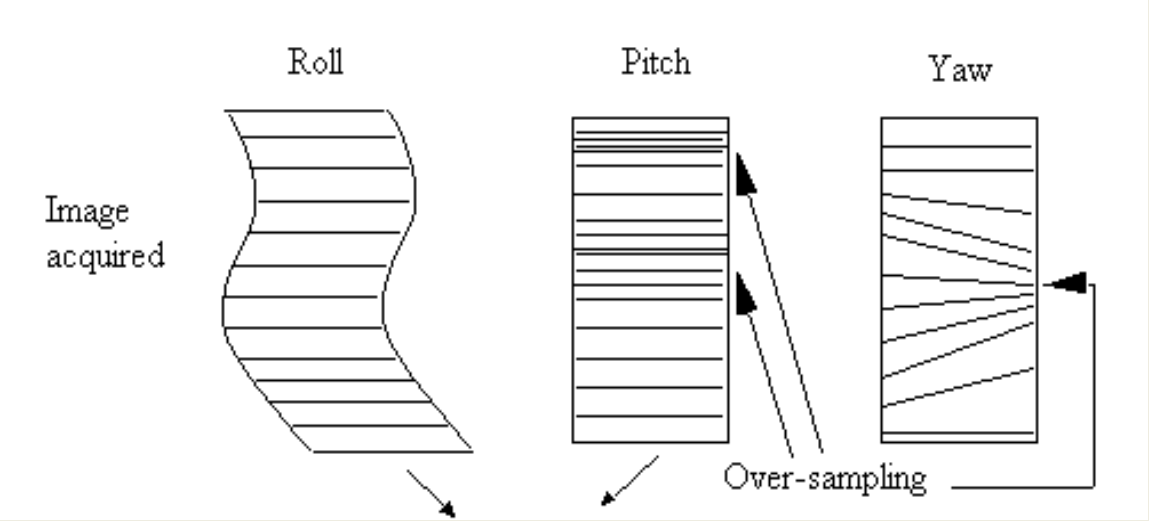


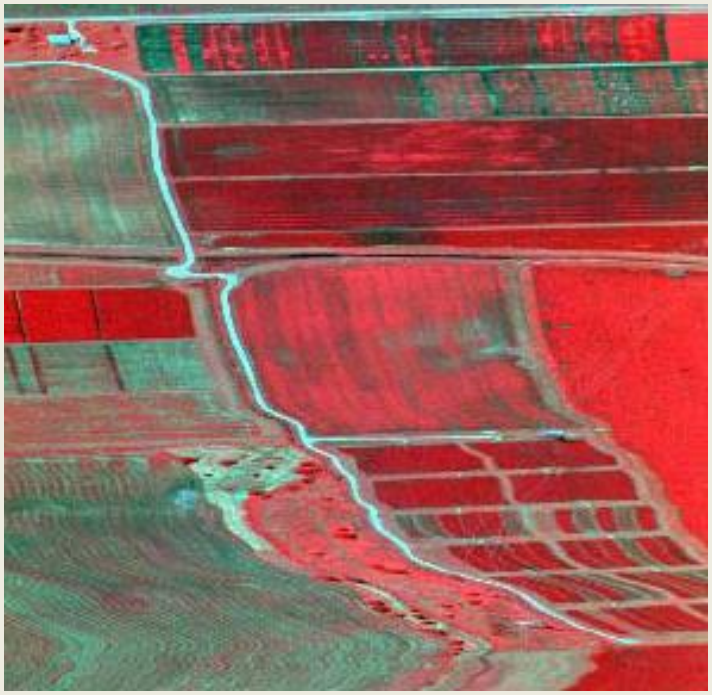
AIRCRAFTS

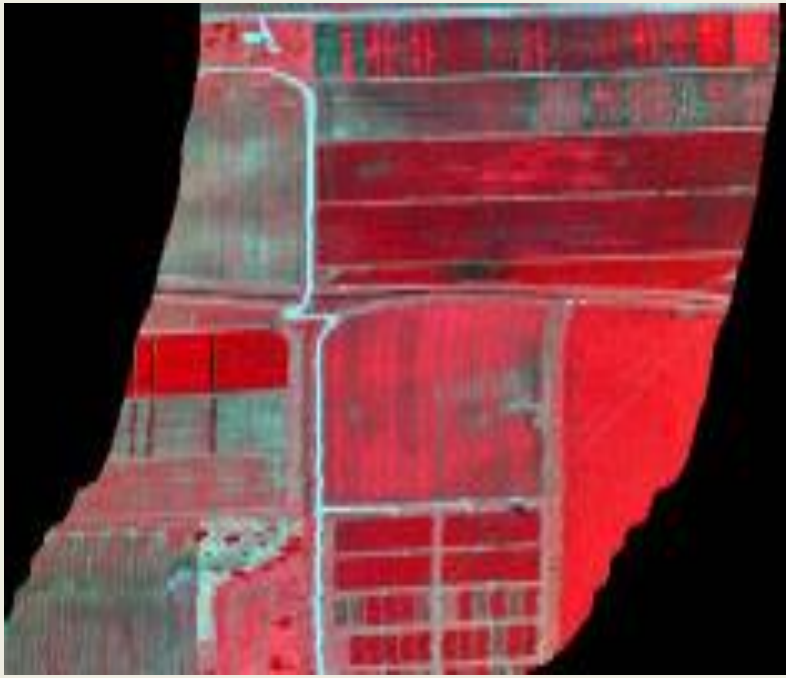

+
0
-
roll angle
+
0
-
pitch angle
+
0
-
yaw angle


Roll
Pitch
Yaw
Image
acquired
Over-sampling

AIRCRAFTS











UAV/Drones











Figure 2: G180 – fixed wing UAV.

UAV/Drones

- Quick Image acquisition
- Operation in confined areas (Urban)
- Relatively very low cost data
- High resolution image
- Not limited to the lengthy image acquisition planning and coordination

UAV/Drones

- Less weather dependency- flying altitude can be adjusted below the cloud
- Operational in hostile environment
- In case of emergency, rain, snow, fog, or rough terrain will not stop UAS from obtaining valuable geographical information.
- There will be safety concern if manned aircraft is deployed

UAV/Drones

- **Limitations –**
  - **Obtaining permission,** Regulatory challenges- emanating from safety, security and privacy concerns
  - **Very smaller area coverage**
  - **Less Payload**
  - **Low Altitude**
  - **Less flight time, power issues**
  - **Weather Dependence**
  - **Possibility of systematic distortions (roll, pitch, yaw)**
  - **Security reasons**

SPACECRAFTS

- MANNED
  - Soviet's Cosmonauts
  - Gemini, Apollo
  - Space shuttle
- UNMANNED
  - TIROS-1
  - LANDSAT
  - SPOT, SENTINEL etc



ADVANTAGES-UNMANNED

- **Do not require complex and heavy life support system**
- **Possibility of complete global coverage** (whole earth)
- **Can operate for very long periods subject to deterioration of equipment**
- **Repeated coverage**
- **Large area coverage at lower cost**

DISADVANTAGES-UNMANNED

- Coverage limited to fixed orbital path
- Fix repeat cycle
- Atmospheric effects
- Cloud Coverage

MISSION CLASSIFICATION

- **Earth resource imaging missions**
  - **LANDSAT, SPOT, IKONOS**
- **Planetary missions (Imaging of other celestial objects)**
  - **HUBBLE SPACE TELESCOPE, APPOLO**

